

Work Breakdown Structure — Version 1

Project: Snacks Stocks Management System **Team 1:** Han, Nhu, Ricky, Hoang **Period:** 2026-03-24 → 2026-04-30 (Unified Process Inception + Elaboration + Construction) **Last updated:** 2026-04-18

Follows WBS guidelines from OOSE2026Discussion0414 (slide 43): each element uniquely identified, uniform work type, minimal dependencies, well-defined completion criteria.

1. Project Management

ID	Task	Owner	Est. effort	Completion criterion	Status
1.1	eCourse2 team space set-up	Hoang	0.5 d	Space created + shared	✓
1.2	Google Drive materials folder	Hoang	0.5 d	Folder link in chat	✓
1.3	WBS + schedule publication	Ricky	1 d	This file on eCourse2	🔄
1.4	Weekly stand-up memos	All (rotating)	0.25 d × N	Memo uploaded per class	🔄

2. Requirements — SRS (due 2026-04-28)

ID	Task	Owner	Est.	Completion	Status
2.1	Skeleton (IEEE 830-1998)	Ricky	0.5 d	docs/SRS.md committed	✓
2.2	Fill Problem / Scope / Users	Ricky	0.5 d	§1, §2 complete	⌚
2.3	Functional requirements F1–F5	Ricky + Han review	1.5 d	§3 complete	⌚
2.4	Non-functional requirements	Ricky	0.5 d	§5 complete	⌚

ID	Task	Owner	Est.	Completion	Status
2.5	Review + sign-off	All	0.5 d	All team  on doc	

3. Design — SDD (due 2026-04-28)

ID	Task	Owner	Est.	Completion	Status
3.1	Skeleton (IEEE 1016-2009)	Ricky	0.5 d	docs/SDD.md committed	
3.2	Use-case diagram	Ricky	0.5 d	PNG/PlantUML in docs/uml/	
3.3	Class diagram	Ricky + Nhu	1 d	PNG/PlantUML in docs/uml/	
3.4	Sequence diagrams (consume / restock / alert)	Ricky	1 d	3 diagrams in docs/uml/	
3.5	State diagram (SnackItem)	Ricky	0.5 d	Diagram in docs/uml/	
3.6	Architecture view (MVC + Observer)	Ricky	0.5 d	§5.2 + diagram	
3.7	Review + sign-off	All	0.5 d	All team 	

4. Implementation (due 2026-04-30 demo)

ID	Task	Owner	Est.	Completion	Status
4.1	Stack decision + repo scaffold	Nhu	0.5 d	Repo + README	
4.2	Data model migration	Nhu	1 d	Schema matches SDD §5.4	
4.3	Inventory CRUD (F1)	Nhu	2 d	Manual test green	
4.4	Expiration tracking + Observer (F2)	Nhu + Han	1.5 d	Dashboard updates on state change	
4.5	Consumption / Restock logging (F3)	Nhu	1 d	Immutable audit entries	
4.6	Dashboard (F4)	Nhu	1 d	Matches Figma	

ID	Task	Owner	Est.	Completion	Status
4.7	Figma → React/HTML components	Nhu	parallel	UI matches design	

5. Testing (bonus credit, due 2026-04-28)

ID	Task	Owner	Est.	Completion	Status
5.1	Test plan (ISO/IEC/IEEE 29119-3)	Ricky	0.5 d	<code>docs/TestPlan.md</code>	
5.2	Test specifications for F1–F4	Ricky	1 d	Cases ID'd per requirement	
5.3	Unit tests on Model layer	Nhu	1 d	≥ 70 % line coverage	
5.4	Test execution log + report	Ricky	0.5 d	<code>docs/TestReport.md</code>	

6. Pattern Sales (due 2026-04-28, 15-min presentation)

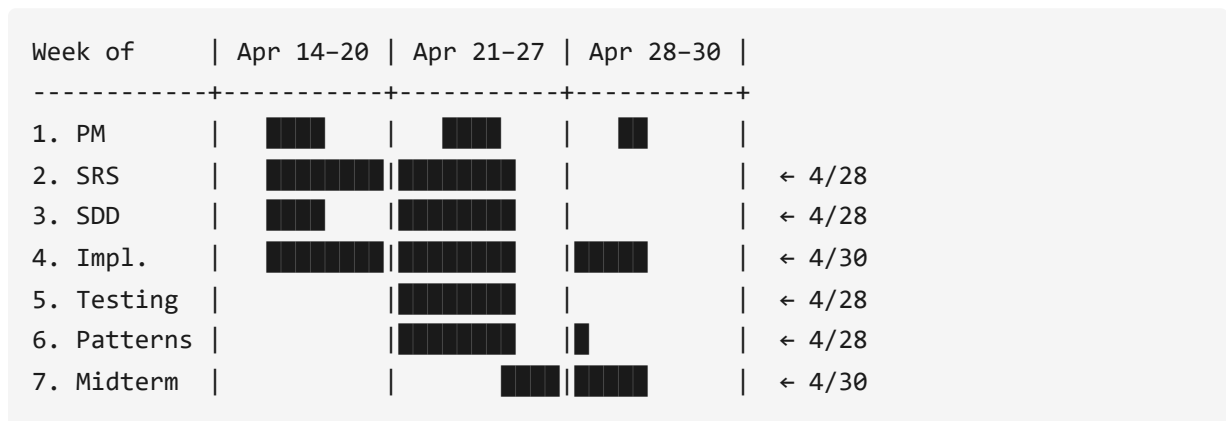
ID	Task	Owner	Est.	Completion	Status
6.1	Sign-up for Decorator / Iterator / CQRS	Han	0.25 d	Listed on eCourse2	
6.2	Decorator slides + code demo	Hoang	1 d	Ppt + runnable snippet	
6.3	Iterator slides + code demo	Hoang	0.5 d	Ppt + runnable snippet	
6.4	CQRS (architecture) slides + project tie-in	Han	1 d	Ppt + tie-in to Snacks Stocks design	
6.5	Rehearsal	Han + Hoang	0.5 d	15-min run-through on time	

7. Midterm Report & Demo (due 2026-04-30)

ID	Task	Owner	Est.	Completion	Status
7.1	Update <code>v1.pptx</code> with final arch / use cases	Ricky	1 d	New deck committed	

ID	Task	Owner	Est.	Completion	Status
7.2	Prepare demo script (happy path + expiry alert)	Nhu	0.5 d	Script in <code>docs/DemoScript.md</code>	
7.3	Dry-run full demo	All	0.5 d	End-to-end runs clean	
7.4	Submit midterm doc bundle on eCourse2	Ricky	0.25 d	Uploaded before 23:59	

Schedule Overview (Gantt-style)



Critical Path

SDD skeleton → Class & sequence diagrams → Implementation → Demo dry-run If the class diagram slips past 2026-04-23, coding risk is high.

Risks

#	Risk	Likelihood	Mitigation
R1	Stack not chosen by 4/20 blocks coding	High	Ricky to poll Nhu today; pick default (Flask + SQLite) if no decision
R2	SRS gaps discovered late	Medium	Review on 4/23
R3	Pattern Sales code demo clashes with coding timeline	Medium	Reuse project codebase snippets
R4	Team coordination looseness	High	Daily 10-min check-in in group chat (no video needed)